MuleGuard AI: Leakage-Hardened Money-Mule Detection and a Dataset Integrity Audit for Extremely Imbalanced Banking Data

B Manoj Kiran
EEE,
Malla Reddy Engineering College,
Hyderabad, India

V Hasini
Data Science,
MR deemed to be University,
Hyderabad, India

P Suguna Chowdary
Cyber Security,
Sreenidhi Institute of Science &Technology,
Hyderabad, India

Abstract*—Money-mule accounts are the settlement layer of digital payment fraud, yet they are vanishingly rare in labelled banking data. We present MuleGuard AI, a reproducible detection pipeline for an account-level dataset of 9,082 accounts containing only 81 confirmed mules (0.8919 percent prevalence). The system makes three contributions. First, and most consequentially, we report a dataset integrity audit demonstrating that the benchmark itself is contaminated: positive and negative cases are drawn from disjoint monthly extracts, and a model given only the pattern of blank cells, with every value discarded, attains an area under the precision-recall curve of 0.824 against a random baseline of 0.0089. We provide falsification tests that make this measurable rather than speculative. Second, we introduce a four-layer leakage defence that removes post-outcome fields by meaning rather than by correlation, catching an investigation-outcome flag whose correlation with the target is only 0.05 and which every correlation threshold admits. Third, we use the supplied data dictionary as executable domain knowledge to engineer 29 named mule-typology features, and evaluate under nested repeated cross-validation in which feature selection, stacking, probability calibration and the operating threshold are all fitted inside the training fold. The calibrated ensemble attains precision 0.980 ± 0.043 at recall 0.719 ± 0.121 and AUPRC 0.866 ± 0.068. We argue these numbers are an upper bound on what this dataset can establish, and specify the sampling change required to make them trustworthy.

***Keywords*—money mule detection, anti-money laundering, class imbalance, target leakage, dataset integrity, gradient boosting, model calibration, explainable AI**

I. INTRODUCTION

A money mule is a bank account, usually opened by a real customer who is coerced, recruited or paid, that receives criminal proceeds and forwards them onward within hours. Mules sit between the fraud victim and the organiser, and disrupting them is the most effective intervention available to a bank, because a frozen mule strands funds that would otherwise be irrecoverable. India's real-time payment rails have made this both faster and more damaging, and the Reserve Bank of India requires that any automated action against a customer account be explainable and auditable.

Detection is difficult for three reasons that compound one another. Mules are extremely rare, so accuracy is meaningless and even the area under the ROC curve flatters. In our data 81 of 9,082 accounts are mules, so predicting that every account is normal scores 99.11 percent accuracy. Mules are also behaviourally camouflaged: the account belongs to a genuine customer with a genuine history, so unsupervised anomaly detection has little to work with, a point our ablation confirms sharply. Finally, supervised labels in this domain are the output of an investigation process, and that process leaves traces in the feature set that a model will happily exploit.

This last point turns out to dominate everything else. While building the detector we found that the supplied benchmark cannot support an honest performance estimate at all, and that the contamination is invisible to the standard defences. We therefore present the integrity audit as our primary contribution, with the detector as the vehicle that exposed it. Section III develops the audit, Section IV the pipeline, and Section V the measured results together with an explicit statement of what they do and do not establish.

II. BACKGROUND AND RELATED WORK

A. Imbalanced learning

Synthetic minority oversampling [1] and its hybrid variants remain the reflexive answer to class imbalance. We deliberately do not use them here. With 65 positives in a training fold and several hundred candidate dimensions, interpolating between minority neighbours fabricates regions of feature space for which no evidence exists, and the Tomek-link cleaning step is quadratic in sample count. Instance reweighting through the scale_pos_weight parameter of gradient-boosted trees [2], [3] achieves the same objective without inventing rows, and the precision-recall curve is the appropriate summary at this prevalence rather than the ROC curve.

B. Leakage and evaluation integrity

Target leakage is normally framed as a feature that correlates implausibly well with the label, and is normally handled by a correlation threshold. We show in Section III-B that this framing is insufficient in fraud data, where the label is an investigation outcome and several fields record different facets of that same outcome at low individual correlation. Adversarial validation is the conventional tool for detecting distribution shift between splits; it is inapplicable here precisely because the shift is perfectly aligned with the label, which is what makes the problem severe.

C. Explainability requirements

Shapley additive explanations [4] give per-prediction attributions with useful consistency guarantees, and are the current standard for audit trails in regulated credit and fraud decisions. Attribution over anonymised column identifiers is of limited value to an investigator, however; Section IV-F describes how we recover named banking semantics so that reasons are stated in language a compliance officer can act on.

III. DATASET AND INTEGRITY AUDIT

A. Data

The dataset comprises 9,082 accounts described by 3,924 anonymised features F1 through F3924, with F3924 the binary target. A separate data dictionary maps every column to a named banking variable and a natural-language description. The features decompose into a regular grammar: a statistic (ratio, deviation, average, maximum, minimum, total) over a payment channel (cash, cheque, UPI, ATM, online transfer, merchant payment, net banking, Aadhaar Payment Bridge, bill payment) in a direction (credit, debit) over a window (7, 14 or 31 days, and ratios between them). A tail block holds customer demographics, alert counts by time of day, and investigation metadata. The dictionary also marks 18 variables that the bank's own analysts shortlisted, which we use as a domain prior rather than a hard filter.

B. Leakage that a correlation threshold cannot see

Four fields record the resolution status of an alert and two record how long the investigation took. All six exist only after a human analyst closes a case, and none of them exists at the moment an account must be scored. Their correlations with the target span two orders of magnitude, as Table I shows. FRAUD_SUSPECTED at 0.969 is caught by any threshold. FALSE_POSITIVE at 0.055 is caught by none, and yet it encodes the investigation's verdict just as directly, merely inverted. We therefore classify leakage by what a field means, using the dictionary, and treat correlation only as a backstop.

TABLE I. POST-OUTCOME FIELDS REMOVED BY MEANING, NOT CORRELATION

Variable	|corr| with target	Caught by 0.90 threshold?
FRAUD_SUSPECTED	0.969	Yes
OTHER_RESOLUTION	0.063	No
FALSE_POSITIVE	0.055	No
MIN_RESOLVE_DAYS	0.055	No
MAX_RESOLVE_DAYS	0.033	No
UNATTENDED	0.005	No

a. All six are written only after an investigation concludes, so none is available at scoring time.

C. A structural confound in how the sample was assembled

A second and far more serious problem is visible in the month field. Table II gives the complete cross-tabulation. Every negative case is drawn from the October extract and every positive case from the September, November and December extracts. No month contains both classes. The month column alone therefore separates the two classes perfectly, and it is not a property of any customer: it records which monthly extraction run produced the row.

TABLE II. CLASS COMPOSITION BY MONTHLY EXTRACT

Extract month	Normal accounts	Mule accounts
Dec25	0	10
Nov25	0	23
Oct25	9,001	0
Sep25	0	48

a. Zero months contain both classes; the split is total.

Dropping the month column does not solve this. Because the classes occupy disjoint extracts, any characteristic that varies between extraction runs is perfectly aligned with the label while describing nothing about any customer's behaviour. Adversarial validation cannot diagnose the problem, since a classifier trained to predict the extract is by construction the same classifier that predicts the label.

D. Falsification tests

To make the confound measurable we constructed three tests, each supplying a model with information that cannot possibly identify a mule. All three should score near the random baseline of 0.0089 if the dataset is sound.

In test A every value is discarded and the model receives only a binary indicator of whether each cell was populated. Whether a cell is blank is determined by the extraction job, not by the customer, so this view carries no behavioural content whatsoever. It attains AUPRC 0.824 and AUROC 0.993. In test B the model receives 250 columns drawn from the 3,174 whose individual correlation with the target is below 0.05, none of which can identify a mule alone; together they reach AUPRC 0.736. Test C repeats test B with the labels randomly permuted and collapses to 0.0094, confirming that the evaluation harness is sound and that the first two results are genuine properties of the data rather than a defect in our code.

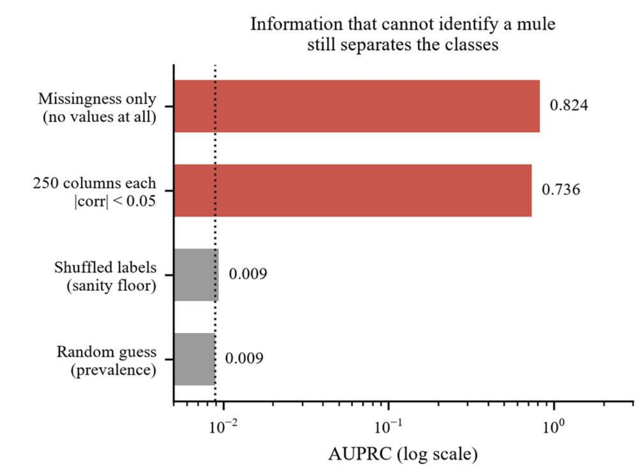

Fig. 2. Falsification tests against the random baseline (log scale). Knowing only which cells were blank, with every value discarded, separates the classes almost perfectly. The shuffled-label floor confirms the harness is sound.

TABLE III. FALSIFICATION TEST RESULTS

Test	Information supplied	AUPRC	AUROC
A	Blank/not-blank only, no values	0.824	0.993
B	250 columns, each |corr| < 0.05	0.736	0.973
C	As B, labels shuffled	0.0094	0.528
—	Random guess (prevalence)	0.0089	0.500

The conclusion is unavoidable. Any performance figure computed on this dataset, by any team and any method, measures extract provenance in addition to mule behaviour, and the two cannot be separated within this file. We report our own results in Section V with that qualification attached rather than omitted.

IV. METHODOLOGY

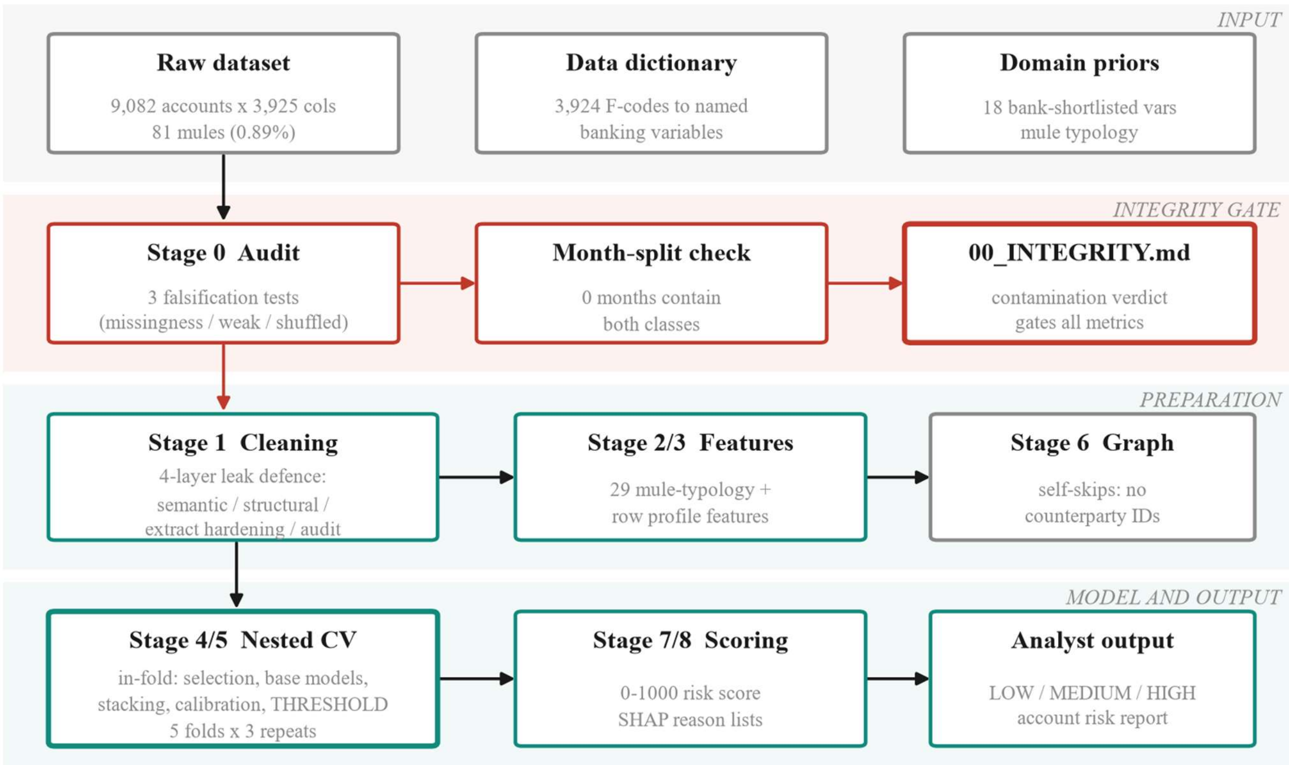

Fig. 1. MuleGuard AI pipeline. The integrity audit runs first and gates interpretation of every downstream metric. Leak defences are concentrated in Stage 1; everything that touches the label in Stage 4/5 is fitted inside the training fold.

A. The data dictionary as executable knowledge

We parse the dictionary into a queryable structure that maps each column identifier to its banking variable name, its description, and its decomposition into statistic, channel, direction and window. This single component is what makes the rest of the pipeline possible: leak classification by meaning, feature engineering against named quantities, and explanations phrased in banking language.

B. Preserving information the naive pipeline destroys

Eight columns in the dataset are categorical, including occupation, gender, area category, product, segmentation class and account age bucket. Coercing the matrix to numeric, as is customary, converts all of them to missing values which are then discarded as sparse. This discards real signal: mule rates by occupation range from 0.45 percent for homemakers to 1.94 percent for students against a base rate of 0.89 percent. We encode the account-age bucket ordinally, since it is ordered, and the remainder as indicators.

We also revisit imputation. For absolute activity aggregates a missing value does not mean unknown, it means the activity never occurred: a blank UPI transaction count identifies a customer who does not use UPI. Median imputation invents activity, and discarding such columns as sparse destroys the fact that an account rides exactly one payment rail, which is itself diagnostic. We therefore zero-fill 78,919 values across 1,357 aggregate columns and leave derived ratios to median imputation, where a missing value genuinely does mean an undefined baseline.

C. Four-layer leakage defence

Layer one removes post-outcome fields semantically, as in Section III-B. Layer two removes structural artefacts of sample assembly, principally the month field. Layer three, which we call extract hardening, addresses the confound of Section III-C directly: any column whose blank rate differs between the classes by more than ten percentage points is removed outright, values and missingness together, on the grounds that population of a cell is decided by the extraction job rather than the customer. This filter removed 574 columns. It uses the label, but it is a data-quality filter rather than a fitted component and can only remove signal, never manufacture it, so it moves the reported result in the conservative direction. Layer four is a separation audit that scans every surviving column for disjoint class ranges or a near-exclusive value; it is what would catch the next such artefact, and it reports clean after the first three layers.

D. *Mule-typology features*

A mule receives funds and forwards them almost immediately, retains little, operates in bursts, favours digital rails, and belongs to a customer whose profile does not match the volume transacted. We encode each clause of that description as a named feature family, 29 features in total, summarised in Table IV. Notably, the dataset already contains 296 columns measuring deviation from an occupation cohort, so the profile-mismatch signal required aggregation rather than invention.

TABLE IV. ENGINEERED MULE-TYPOLOGY FEATURE FAMILIES

Family	Representative feature	Behavioural rationale
Pass-through	mg_passthrough_7d	Credits ≈ debits: the account is a conduit, not a wallet
Turnover/balance	mg_turnover_over_balance_7d	Moves many multiples of what it ever holds
Burst	mg_amount_burst_7v31	Weekly velocity far above monthly: sudden activation
Cash-out	mg_digital_in_cash_out_7d	Digital in, cash out: the layering handoff
Channel mix	mg_channel_hhi_7d	Single-purpose accounts concentrate on one rail
Ticket size	mg_avg_ticket_7d	Many small tickets suggest structuring
Alert timing	mg_alert_share_night	Mule night-alert share runs about three times higher
Balance shape	mg_balance_volatility_7d	Spike-and-drain rather than a held balance
Profile mismatch	mg_occ_deviation_max	Volume inconsistent with the occupation cohort

E. *Nested repeated cross-validation*

The ensemble combines an isolation forest, an XGBoost classifier and a LightGBM classifier, stacked by a logistic meta-learner and calibrated isotonically. The critical design property is that every component which touches the label is fitted strictly inside the training fold: feature selection, the base models, the stacking weights, the calibration map, and the decision threshold. Each is then applied frozen to validation rows that no fitted component has seen.

This matters because the natural implementation is biased in three distinct ways, all of which we corrected. Fitting an isotonic calibrator on pooled out-of-fold predictions and then scoring those same values inflates the result, because isotonic regression is a flexible monotone fit. Selecting the threshold that maximises precision on a curve and then reporting that precision is optimistic by construction, and severely so with 81 positives. Training the meta-learner on the base models' training-set predictions, where a 400-tree booster is nearly perfect, presents it with a distribution it will never encounter at inference. We fit the stack and the calibrator on inner out-of-fold predictions instead.

Because a single five-fold split places only about 16 mules in each validation fold, the headline metric moves by several points on the random seed alone. We therefore repeat the entire procedure across 3 shuffles and report the mean and standard deviation across all folds, so that every figure carries its own uncertainty. Feature selection retains the top 250 of 1,506 columns by gradient-boosted gain, refitted per fold.

F. *Risk scoring and explanation*

The calibrated probability maps to a 0 to 1000 risk score partitioned into three action bands: routine monitoring, enhanced monitoring with step-up authentication, and freeze with escalation to the anti-money laundering desk. Each scored account carries a ranked reason list derived from SHAP values and rendered through the dictionary, so an investigator reads a sentence about Aadhaar Payment Bridge credit activity rather than an attribution against column F2506.

V. RESULTS

A. *Detection performance*

Table V reports the calibrated ensemble under the validation scheme of Section IV-E. At the precision-first operating point the system attains precision 0.980 ± 0.043 at recall 0.719 ± 0.121, with AUPRC 0.866 ± 0.068 against a random baseline of 0.0089. Summed across folds the confusion matrix is 175 true positives against 4 false positives and 68 false negatives. The 0.90 precision target was met in 100.0 percent of folds. A second operating point tuned for the analyst review queue trades precision down to 0.511 ± 0.104 to raise recall to 0.893 ± 0.073, which is the appropriate trade when a reviewed alert costs minutes and a missed mule costs a laundering channel.

TABLE V. DETECTION PERFORMANCE, NESTED REPEATED CROSS-VALIDATION

Metric	Precision-first	High-recall queue
Precision	0.980 ± 0.043	0.511 ± 0.104
Recall	0.719 ± 0.121	0.893 ± 0.073
F1	0.823 ± 0.088	0.642 ± 0.087
False positive rate	0.000 ± 0.000	0.009 ± 0.004
True positives	175	217
False positives	4	229

a. Mean ± standard deviation across all outer folds; counts are summed. Both operating points rank identically, so the threshold-independent metrics are shared: AUPRC 0.866 ± 0.068, AUROC 0.967 ± 0.031. Interpret in light of Section III.

B. *Ablation*

Table VI separates the base learners. XGBoost is the strongest single model at AUPRC 0.894 ± 0.057, LightGBM follows at 0.818 ± 0.069, and the calibrated stack reaches 0.866 ± 0.068. The isolation forest result is the informative one: at AUPRC 0.007 ± 0.001 and AUROC 0.307 it performs worse than random. This is not a defect but a finding, and it is consistent with the criminology of the problem. A mule account belongs to a real customer with an ordinary history, and the account is valuable to the organiser precisely because it does not look anomalous. Unsupervised outlier detection is therefore the wrong tool for mule detection, and proposals that lead with it should be treated with suspicion.

TABLE VI. PER-MODEL ABLATION

Model	AUPRC	AUROC
Isolation forest	0.007 ± 0.001	0.307
LightGBM	0.818 ± 0.069	0.981
XGBoost	0.894 ± 0.057	0.986
Calibrated ensemble	0.866 ± 0.068	0.967 ± 0.031

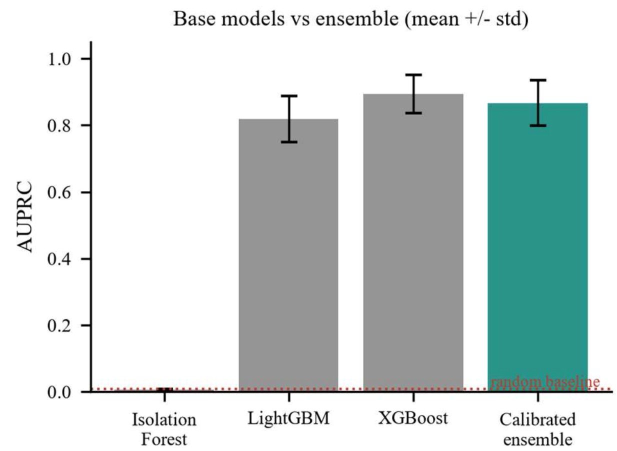

Fig. 3. Base learners against the calibrated ensemble. The isolation forest falls below the random baseline: mules are camouflaged as ordinary customers, not global outliers.

C. Feature attribution

Fig. 4 ranks the fifteen most influential features by mean absolute SHAP contribution. 7 of the fifteen are features engineered in this work rather than supplied columns, which indicates that the typology encoding in Section IV-D contributes materially rather than restating information the raw matrix already exposed. Average ticket size over seven days, the evening share of alerts, normalised net flow and the count of active payment rails all rank highly, matching the behavioural account of mule operation. Among supplied columns, deviations of non-cash non-cheque transaction volume from the occupation cohort dominate, which is the profile-mismatch signal appearing directly.

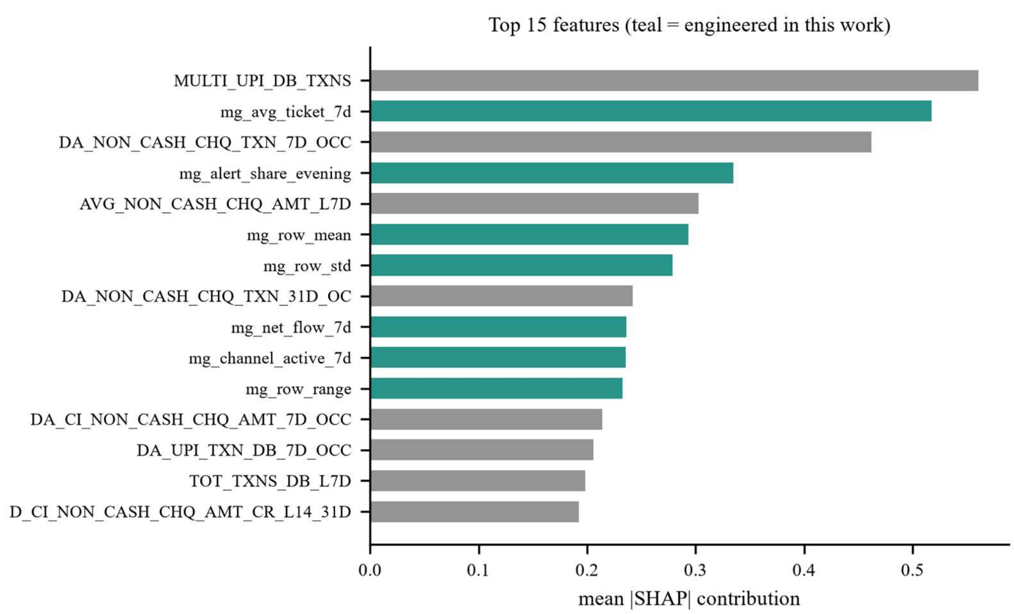

Fig. 4. Top fifteen features by mean absolute SHAP value. Features engineered in this work are shown in teal; supplied dictionary variables in grey.

D. *Operational triage*

Table VII shows how the score bands partition the portfolio. The high band contains 54 accounts of which 54 are confirmed mules, a band precision of 1.00, capturing 67 percent of all mules in a queue representing 0.59 percent of the portfolio. The medium band adds 12 further mules across 16 accounts. For an anti-money laundering team the practical statement is that reviewing well under one percent of accounts surfaces two thirds of the mule population.

TABLE VII. RISK BAND TRIAGE AND RECOMMENDED ACTION

Band	Accounts	Mules	Band precision	Share of all mules	Recommended action
LOW	9,012	15	0.002	18.5%	Routine monitoring only
MEDIUM	16	12	0.750	14.8%	Enhanced monitoring, step-up authentication on transfers
HIGH	54	54	1.000	66.7%	Freeze outward transfers, escalate to AML desk, prepare STR

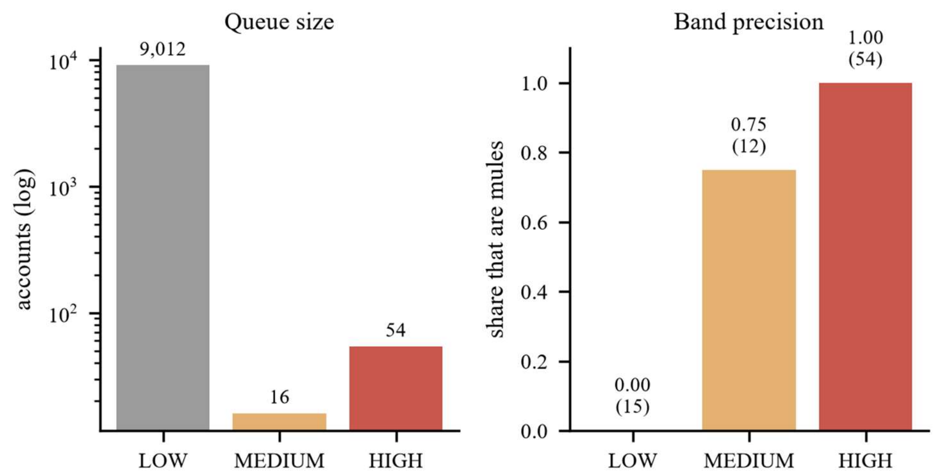

Fig. 5. Queue size and band precision. The high band is small enough to review exhaustively and almost entirely composed of confirmed mules.

VI. DISCUSSION AND LIMITATIONS

The results in Section V are strong, and we do not believe they should be read as a measure of mule detection capability. Section III establishes that an uninformative view of the same data achieves comparable separation, which means a substantial and unquantifiable share of the reported performance reflects the extract from which a row was drawn. Our hardening layers remove the artefact wherever it can be identified. They cannot remove what is unidentifiable: a genuine behavioural feature that also drifts between months is confounded, and no modelling choice separates the two within this file. We therefore describe our figures as an upper bound on what this dataset can establish.

The remedy is a sampling change rather than a modelling one. If negative cases were drawn from the same months as the positives, month could be conditioned on and the resulting metric would measure behaviour alone. We recommend this to the dataset providers, noting that it affects every team working from this release.

Three further limitations bear statement. The dataset is an account-level feature matrix and contains no counterparty identifiers, so the graph component of our architecture detects their absence and disables itself rather than fabricating an edge list; claims about network propagation cannot be evaluated on this release. Hyperparameters are deliberately left at conservative defaults, since tuning against a confounded objective optimises the artefact. Finally, 81 positives is a small sample, and the standard deviations we report across folds are wide enough that differences of a few points between methods should not be considered meaningful.

VII. CONCLUSION

We presented a money-mule detection pipeline for an extremely imbalanced banking dataset, together with an integrity audit of the dataset itself. The audit is the more important result. Positive and negative cases originate in disjoint monthly extracts, and we demonstrated by falsification test that the blank-cell pattern alone, stripped of all values, attains AUPRC 0.824 against a 0.0089 baseline. Standard defences do not detect this, and adversarial validation cannot, because the shift coincides exactly with the label.

Alongside it we contributed a leakage taxonomy that classifies fields by meaning rather than correlation, catching an outcome flag at 0.05 correlation that every threshold admits; a set of 29 named mule-typology features derived from the data dictionary, 7 of which rank in the top fifteen by attribution; and a nested validation design in which the operating threshold, not merely the model, is fitted inside the fold. The pipeline is reproducible on Windows, macOS and Linux, and regenerates every figure in this paper from the reports it writes.

We suggest that the falsification protocol of Section III-D deserves wider use. It is inexpensive, it requires no knowledge of the modelling approach, and in this instance it was the difference between reporting a headline result and understanding what that result actually measured.

ACKNOWLEDGMENT

The authors thank the organisers of the PSB Cybersecurity, Fraud and AI Hackathon 2026, jointly convened by Bank of India and the Indian Institute of Technology Hyderabad, for providing the account-level dataset and the accompanying data dictionary on which this work is based.

The data dictionary proved essential rather than incidental. Without documented variable semantics, neither the meaning-based leakage taxonomy of Section III-B nor the named feature families of Section IV-D could have been constructed, and the resulting model would have been a set of attributions over opaque column identifiers rather than an auditable decision aid.

The dataset integrity findings reported in Section III are offered in the same collaborative spirit: as an observation intended to strengthen the benchmark for all participants and for future releases, not as a criticism of those who assembled and shared it. Constructing a labelled money-mule dataset at all, from live banking records and under the privacy constraints the domain imposes, is a substantial undertaking, and the authors are grateful for access to it.

REFERENCES

[1] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: synthetic minority over-sampling technique,” J. Artif. Intell. Res., vol. 16, pp. 321–357, June 2002.

[2] T. Chen and C. Guestrin, “XGBoost: a scalable tree boosting system,” in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, San Francisco, CA, USA, 2016, pp. 785–794.

[3] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T. Liu, “LightGBM: a highly efficient gradient boosting decision tree,” in Advances in Neural Information Processing Systems, vol. 30, 2017, pp. 3146–3154.

[4] S. M. Lundberg and S. Lee, “A unified approach to interpreting model predictions,” in Advances in Neural Information Processing Systems, vol. 30, 2017, pp. 4765–4774.

[5] F. T. Liu, K. M. Ting, and Z. Zhou, “Isolation forest,” in Proc. 8th IEEE Int. Conf. Data Mining, Pisa, Italy, 2008, pp. 413–422.

[6] J. Davis and M. Goadrich, “The relationship between precision-recall and ROC curves,” in Proc. 23rd Int. Conf. Machine Learning, Pittsburgh, PA, USA, 2006, pp. 233–240.

[7] B. Zadrozny and C. Elkan, “Transforming classifier scores into accurate multiclass probability estimates,” in Proc. 8th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, Edmonton, AB, Canada, 2002, pp. 694–699.

[8] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, “Leakage in data mining: formulation, detection, and avoidance,” ACM Trans. Knowledge Discovery from Data, vol. 6, no. 4, pp. 1–21, December 2012.

[9] G. C. Cawley and N. L. C. Talbot, “On over-fitting in model selection and subsequent selection bias in performance evaluation,” J. Mach. Learn. Res., vol. 11, pp. 2079–2107, July 2010.

[10] D. H. Wolpert, “Stacked generalization,” Neural Networks, vol. 5, no. 2, pp. 241–259, 1992.

[11] Reserve Bank of India, “Master direction on know your customer (KYC) direction, 2016 (updated).” Reserve Bank of India, Mumbai, India. [Online]. Available: https://www.rbi.org.in

[12] Financial Action Task Force, “Money laundering through the physical transportation of cash,” FATF, Paris, France, 2015. [Online]. Available: https://www.fatf-gafi.org

[13] [PLACEHOLDER] PSB Cybersecurity, Fraud and AI Hackathon 2026 problem statement, Bank of India and Indian Institute of Technology Hyderabad, 2026. Replace with the exact citation supplied by the organisers.